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Opgave: $f(x, y, z) = e^{xyz}$ in $a(4, 0, -3)$ in de richting $h(0, 1, -1)$

$$\begin{aligned} Df(a, h) &= \lim_{\lambda \rightarrow 0} \frac{f(a + \lambda h) - f(a)}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{f((4, 0, -3) + \lambda(0, 1, -1)) - f(4, 0, -3)}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{e^{-4\lambda(3+\lambda)} - e^{4 \cdot 0 \cdot (-3)}}{\lambda} = \lim_{\lambda \rightarrow 0} \frac{e^{-12\lambda-4\lambda^2} - 1}{\lambda} \stackrel{H}{=} \lim_{\lambda \rightarrow 0} \frac{(-12 - 4\lambda)e^{-12\lambda-4\lambda^2}}{1} \\ &= \lim_{\lambda \rightarrow 0} (-12e^{-12\lambda-4\lambda^2} - 4\lambda e^{-12\lambda-4\lambda^2}) = -12 \lim_{\lambda \rightarrow 0} (e^{-12\lambda-4\lambda^2}) - 4 \lim_{\lambda \rightarrow 0} (\lambda e^{-12\lambda-4\lambda^2}) \\ &= -12 \end{aligned}$$

References